Renewable Energy 201 (2022) 150–156

Contents lists available at ScienceDirect

Renewable Energy

journal homepage: www.elsevier.com/locate/renene

Life cycle assessment and circularity evaluation of a PV panel integrated with phase change material

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ARTICLE INFO

Keywords:
Life cycle assessment
Circular economy
Photovoltaic
Phase change materials
Thermal energy storage
Domestic hot water

ABSTRACT

Many studies focus on the application of phase change materials (PCMs) to a photovoltaic (PV) panel. It has been proved that these materials provide a good thermal management for the PV panel, increasing the electrical conversion efficiency and recovering useful heat. This paper investigates these innovative systems from an environmental sustainability perspective. A PV-PCM prototype, built and tested over a reference year, has been taken as case study. The recovered heat by the PCM is intended as peak shaving for the domestic hot water (DHW) energy demand of a typical residential building. The analysis involves both the Life Cycle Assessment (LCA), basing on the ISO standard 14040, and the circularity of the proposed system, proposing an alternative calculation procedure to the new Italian standard UNI 1608856. Results show that the proposed system allows the reduction of impact on climate change of around 11%, reaching a potential circularity performance of around 45%.

1. Introduction

Nowadays daily life is regulated by use of energy for many purposes, such as heating and cooling of buildings, production of goods and services, connections, and transports. The large fossil fuel consumption, which has characterized the last decades, has increased the levels of environmental pollution, soil consumption and other climate-changing factors. Solutions are required, and strategies, strongly based on renewable energy sources (RES), are mandatory to implement. According to Eurostat, the latest statistics show an increase of the RES share, moving from 4.3% in 1990 to 5.3% in 2000 and 8.8% in 2010, and up to 11.8% in 2020 [1]. This positive trend is related to EU's strive to stabilize and reduce greenhouse gas emissions, and to avoid the undesirable and dangerous effects of climate change [2].

In addition, global consumption of materials such as biomass, fossil fuels, metals and minerals is expected to double in the next forty years [3], while annual waste generation is projected to increase by 70% by 2050 [4]. With the aim to reduce this impact, the European Commission (EU) adopted the new circular economy action plan (CEAP) in March 2020. The aim is to promote growth by transitioning to a modern, resource-efficient, and competitive economy [5].

The Sustainable Development Goals are a call for action by all countries to promote prosperity while protecting the planet. The protocol is divided into 17 points. They recognize that ending poverty must go together with strategies that build economic growth and address a range of social needs [6]. Particularly, the goal 12 concerns the "Ensure sustainable consumption and production patterns". In this goal, the 12.5 target focuses on the substantially reduction of waste generation through prevention, reduction, recycling, and reuse by 2030. The target 12.6 encourages companies to adopt sustainable practices and integrate sustainability information into their reporting practice [7].

In addition, Circular Economy (CE) aims to overcome the take-make-dispose linear pattern of production and consumption, proposing a circular system in which the value of products, materials and resources is maintained in the economy as long as possible [8].

The design and study of the life cycle of a product fall largely within the main concepts of the CE and climate change reduction plan. A fundamental instrument to encourage an integrated product policy is the Life Cycle Assessment (LCA) [9], an internationally standardized methodology for quantifying different aspects [10,11], such as the energy, environmental loads and potential impacts to product/process/activity along the entire life cycle [12]. With this aim, a Life Cycle Assessment methodology is applied in agreement with the LCA ISO standard 14040 and 14044:2006. This method aims to improve the

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https://doi.org/10.1016/j.renene.2022.11.076
Received 5 August 2022; Received in revised form 26 October 2022; Accepted 18 November 2022
Available online 19 November 2022
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environmental benefits, trade-offs and areas for achieving improvements from the acquisition of raw materials to the end of life ("from the Cradle to the Grave") [13]. This technique's relevance is mainly due to its innovative approach: evaluating all stages of a production process, in different application fields [14].

Nowadays one of the challenges on this topic is the treatment during the disposal phase of traditional PV panels and the risks related to the hazardous waste that they generate; such as lead, cadmium, chromium and nickel [15]. On the other hand, PV modules contain conventional materials, such as aluminum, copper, glass and critical substances and pure material, such as silver and silicon. As regard the latter, Deng et al. [16] mapped out the four most common approaches for the disposal phase of a PV panel, namely landfill (not sustainable), glass recycling (economically feasible but not widespread due to low volumes or not purity of the material), mechanical recycling and thermal recycling (both require a cost reduction). These materials could be recovered and recycled in the PV sector [17], in line with the principles of the circular economy (CE), which leads to a reduction in resource depletion and the maintenance of the increasing number of solar installations [18–20].

The interest for the circularity aspects is also confirmed by the recent Italian standard UNI 1608856 (under evaluation) which aims at clarifying and giving rules useful for the quantification of circularity of products and organization. Among the tools for the quantification of sustainability and circularity, from February 2022 the commission UNI/CT 057 Circular Economy proposed the project UNI 1608856 entitled "Measurement of circularity - Methods and indicators for measuring circular processes in the organization".

This paper aims also to contribute mainly on the environmental aspect of PCMs, as over the last years many studies are focusing on the energetic performance of these materials. Only few studies are based on a LCA approach and these are related to different applications, such as building envelope [21–23], solar collectors [24] or heat pumps [25] integrated with PCM. These works aim to determine the reduction of the environmental impact by means of the integration of phase change material. Results show that the reduction ranges from 7.7% for ventilated facade [21] to 12/14% for bricks [22] and up to 82% for solar collectors [24]. No specific study is present about PCM-integrated photovoltaic panels. In this sense, among the proposed solutions to increase the electric efficiency of a PV panel, the application of phase change materials (PCMs) allows to recover thermal energy while cooling the panel [26]. A PV-PCM system is characterized by a simultaneous production of thermal and electrical energy [27]. PCMs exploit the solid-liquid phase change to store a large amount of energy, in the form of latent heat [28]. The phase change occurs at a nearly constant temperature, with the advantage of maintaining low temperatures and accordingly the auto-discharge thermal losses [29]. The application of PCM on rear surface of a PV panel allows to recover a share of the incident solar radiation which is not converted into electricity [30]. In addition, by cooling the PV panel increases its electric energy production, as temperature and electric conversion efficiency are inversely proportional [31,32].

This work refers to a photovoltaic panel (PV) integrated with PCMs. The novelty of this paper is the application of the Life Cycle Assessment to an innovative PV-PCM system for thermal storage and intended for the DHW service [33]. No similar studies are present in literature about this specific application. The aim is to prove its environmental sustainability considering the whole life of the innovative system (from production to disposal). In addition, the novelty also involves the proposal of an alternative procedure calculation of a new Italian standard, the UNI 1608856, for the circularity evaluation of these type of systems. This topic is becoming increasingly important over the years.

2. Materials and method

2.1. Description of the physical model

The object of the study is the innovative PV-PCM and water tank system described by Ref. [33], intended as peak shaving for the domestic hot water demand (DHW) of a typical residential building. Fig. 1 shows the integrated PV-PCM panel.

More in detail, the chosen PV panel (of monocrystalline type) is characterized by a nominal power of 40 W and a voltage of 12 V. On the rear surface of the PV panel (64.5 cm × 50 cm), a thin sheet of aluminum is glued and equipped with ten fins, placed parallel and equidistant from each other, transversely to the surface. Fins are immersed in the PCM, acting as conductors of heat, and enhancing the heat transfer rate from the PV panel to the PCM. The PCM is contained in a transparent plexiglass box, which has the same external dimensions of the PV panel (64.5 cm × 50 cm) with a thickness of 3 cm, while the internal ones are 62.5 cm × 48 cm x 2.5 cm. The phase change material chosen is a paraffin wax, the RT35 HC, with a melting temperature of 35 °C. The heat absorbed by the PCM during the heating and the phase change can be extracted by a copper heat exchanger. It consists of ten tubes, with an external and internal diameter of 14 mm and 12 mm respectively, and a length of 55 cm, welded into an inlet and an outlet manifold, which have a length of 45 cm and an external/internal diameter of 16 mm and 14 mm respectively.

The integrated panel is coupled with a water tank by pipes and a pump. Fig. 2 shows the final configuration of the proposed system. The aim was to provide thermal energy for the Domestic Hot Water (DHW) service of a residential building. This energy demand in residential building is usually characterized by a peak in the evening, as it is the typical period of presence and accordingly of energy demand by the occupants. The heat stored by the PCM can be used as peak shaving for the DHW demand. The DHW curves demand have been taken in accordance with the Regulation (EU) No 814/2013. The energy demand was estimated considering a typical four-person household with a DHW consumption of 200 L/day. The temperature of the inlet water is increased by 30 °C. The energy associated to this operation is 2.74 kWh/day.

2.2. Data input

The previous analysis experimentally investigated the yearly performance of the proposed PV-PCM system [33]. The system has been tested for a typical day in a summer and an intermediate month under a solar simulator [34], considering the climate conditions of Ancona (Italy), and for a winter month outdoor, under real conditions. Different tests have been performed varying the size of the water tank. The aim was to investigate the share of the DHW energy demand satisfied by the heat recovered by the PCM and stored in the water tank. Results show that with a storage volume of 100 L case, the energy saving turns out to be 23.7% in a typical summer month, 10.7% during an intermediate month, while in winter the melting process of the PCM has not occurred. Table 1 summarizes the obtained results for the best scenario. The energetic results obtained in this previous study are used as input data for the evaluation of the environmental sustainability of the system.

2.3. The environmental sustainability assessment

2.3.1. Goal and scope

The analysis aims to estimate the possible environmental gain achieved by the integration of phase change materials to a traditional photovoltaic panel, as reference case. The analysis of the environmental sustainability of the integrated panel applies the principles of the Life-cycle Thinking assessing the innovative system considering all the steps of its life. The proposed analysis aims to determine the carbon footprint of the innovative PV-PCM system, using an LCA approach, able to

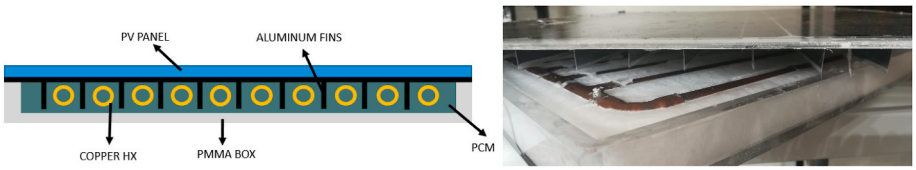


Fig. 1. a- Cross section of the proposed panel integrated with PCM, b- View of the inside of the box [33].

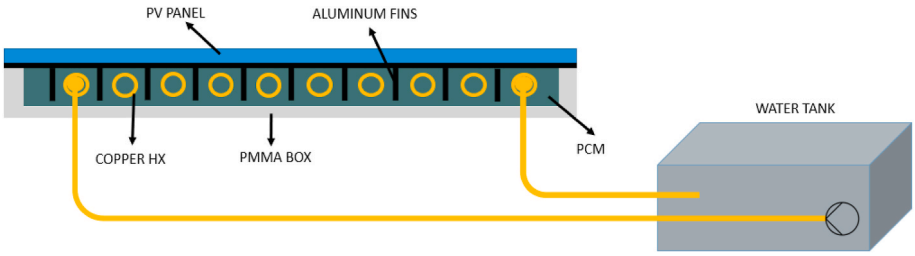


Fig. 2. Final configuration of the proposed system.

Table 1
Results described by Ref. [33].

Season	Energy demand (kWh/day)	Tank Volume (L)	Energy saving (%)	Energy saving (kWh/day)	Energy from grid (kWh/day)
Summer	2.74	100	23.72%	0.650	2.090
Mid months	2.74	100	10.78%	0.295	2.445
Winter	2.74	100	0%	0.000	2.740

includes all the steps of its life (production, use and the end of life). The system boundaries for the analysis are depicted in Fig. 3. The functional unit is the innovative proposed PV-PCM panel, as described in paragraph 2.1, intended for the DHW production of the typical four-person household. The analysis considers both the production and the assembly phase of all materials, and the processes to have the product as finished. The panel life considers a period of 20 years. At the end, the disposal phase differentiates the materials to be recycled and to be intended to the incinerator for energy recovery. The aluminum and the copper of the heat exchanger belong to the first group, while PCM, PMMA, polycarbonate screws and gasket to the second one. In literature, innovative solutions for the recycling of PMMA and PCM are under study, but in this work a conservative approach, waste to energy, has been preferred. The hypothesis of this work is to integrate phase change materials to an existing photovoltaic panel (a common technology widely discussed in the literature), and accordingly only the thermal

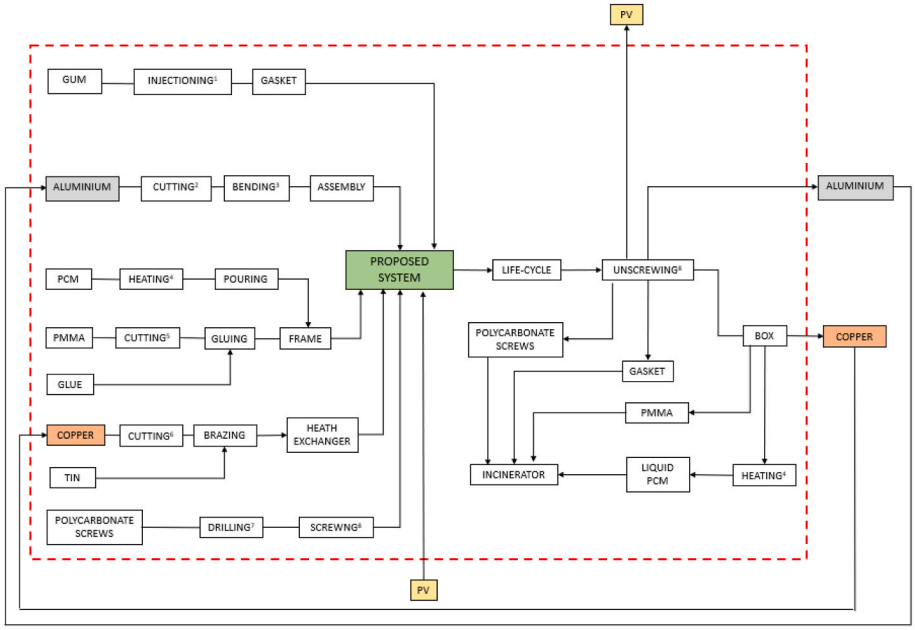


Fig. 3. Boundary conditions of the proposed system (production, life cycle and disposal). [1]Injection molding machine; [2]Mechanical shear; [3]Multi-axis press brake; [4]Electric hotplate; [5]CO_2 Laser Cutting; [6]Automatic cutting center; [7]Column drill; [8]Screwdriver.

contribution of the PCM has been considered. For this reason, the panel has been excluded from the system boundaries in Fig. 3. Similarly, the increase of the electricity production due to the presence of PCM, proved in another work [30], can be considered as a compensation of the energy consumption of the pump. This amount can be considered as a compensation of the energy consumption of the pump, neglected in this work. The same way the impact of the tank has been neglected, as it is already present in the reference scenario (it is assumed that buildings are already equipped with a water tank for the DHW service).

2.3.2. Inventory analysis

In this section a complete inventory is provided, regarding both material and energy flows. Table 2 summarizes all the materials involved in the analysis.

As regard energy flows, Table 3 summarizes the operations details for each component included in the system. The construction phase of fins consists of cutting an aluminum plate with a mechanical shear. Then the pieces have been bended with multi-axis press brake. Subsequently the L-shapes obtained have been glued on a rectangular aluminum plate, with the same dimensions of the PV panel (65 cm × 50 cm). In addition, the copper heat exchanger has been realized by cutting a unique tube with an automatic cutting center to obtain the ten tubes. Then they have been brazed manually using torch cutting. The impact of the brazing phase has been neglected as the consumption of oxygen and acetylene is minimal. The gasket has been obtained by rubber injection in an injection molding machine. The PMMA frame was assembled after CO_2 cutting the pieces from a single plate. Then each element was glued to assemble the frame. Finally, the PCM was heated until complete melting and poured into the PPMA frame. The heat exchanger was already fixed in it, and the aluminum fins with the PV panel were put on as a cover. The whole system was locked by screwing with polycarbonate screws.

The disassembly phase of the proposed system consists of decoupling the upper (PV panel and fins) and the lower (PCM, box and heat exchanger) part. The only operation is unscrewing the polycarbonate screws. Thereafter, each component can be removed.

To evaluate the impact of each operation, the equipment power and the duration of the operation have been considered. The carbon footprint (CF) can be calculated with the following equation (Eq. (1)):

$$CF = \sum_{i}^{n} (P_i \times t_i \times f_i) \quad (\text{kg } CO_2 \text{ eq.}) \tag{1}$$

where P_i (kW) is power of the equipment, t_i (h) is the duration of the operation and f_i (kg CO_2 eq/kWh) is the unitary impact of the energy from the electricity grid used to power the equipment. This way only the impact of the energy associated to the operation is considered, neglecting the impact on the life of the equipment, considered not relevant. The emission factors depend on the energy mix of the local grid. In this case the Italian energy mix has been chosen, whose composition is highlighted in Fig. 4.

For a better completeness and accuracy of the results, the same analysis has been repeated considering Spain and Greece, which have a similar energy production from photovoltaic panels, due to similar weather conditions. In addition, the average European Union energy mix has been also considered. The unitary impacts of the Italian, Spanish, Greek and European Union mixes are 0.45 kg CO_2 eq/kWh, 0.36 kg CO_2 eq/kWh, 0.78 kg CO_2 eq/kWh and 0.372 kg CO_2 eq/kWh, respectively.

Table 2
Composition of the integrated panel.

Material	Weight [kg]
PCM	4.33
Aluminum	0.56
Copper	2.39
Tin	0.05
PMMA	2.56
Gasket	0.23
Polycarbonate screws	0.08

Table 3
Power required by processes and duration.

Process	Equipment power (kW)	Duration(mm:ss)
Aluminum fins		
Mechanical shear	7.5	01:00
Multi-axis press brake	10	05:00
Copper heat exchanger		
Automatic cutting center	8	05:00
Brazing	Negligible	
Gasket		
Injection molding machine	40	02:00
PMMA frame		
CO_2 Laser Cutting	0.85	10:00
Assembling		
Column drill	0.55	20:00
Screwdriver	0.55	05:00
Electric hotplate	1.2	09:00

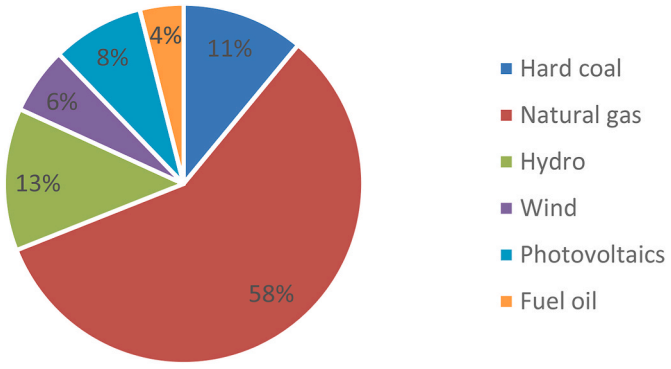


Fig. 4. Italian energy mix.

Furthermore, two different scenarios will be compared:

- Base line scenario: the DHW energy demand is fully satisfied by the national grid, without local production. The system consists of a water tank with an electric immersion heater connected to the grid.
- Proposed system (integrated panel): the thermal energy recovered by the proposed PV-PCM system is intended as peak shaving of the DHW energy demand. The remaining share is provided by the national grid mix.

For each inlet material, the impact on the climate change (CO_2 equivalent) has been assessed. The software used for data collection and elaboration was Thinkstep GaBi 9.2.1.68, combined with the Database for Life Cycle Engineering. The method selected for the analysis (which included the classification and characterization steps) was Environmental Footprint 3.0, including the category of Climate Change. The GaBi software provides the value of impact per unit of mass for a large database of materials. Accordingly, the carbon footprint (CF) is calculated with the following equation (Eq. (2)):

$$CF = \sum_{i}^{n} (m_i \times f_i) \quad (\text{kg } CO_2 \text{ eq.}) \tag{2}$$

where m_i is the mass (kg) of each inlet material and f_i (kg CO_2 eq/kg) is the unitary impact of the material considered.

2.4. Circularity assessment

This work presents an alternative procedure calculation to the Italian standard UNI 1608856 “Measuring circularity - Methods and indicators

for measuring circular processes in organizations". The standard provides a methodology to evaluate the circularity level of a company that produces goods or services. A set of indicators on a 100 basis are defined. Indicators are divided into three categories: mandatory, optional (at least 50% of the indicators has to be considered) and rewarding (that can be chosen freely). Indicators belonging to a group differ with the type of system analyzed, such as a company that deal with products, services, or a mix of them.

In this work the methodology is adapted to the research context, focusing on the innovative product, the PV-PCM integrated panel, rather than for a started company. For this reason, only mandatory indicators which refer to a single product are considered. In this proposed analysis, two different indexes will be proposed. The first one considers the indicators about the circularity of the materials used. Each indicator can be calculated as a ratio between the share of recyclable materials and the totality of the materials. Table 4 shows the details of each indicator. The final index is the average of all indicators calculated.

The second index proposed concerns the energetic performance and the environmental impact of the system under analysis. Two indicators of the mandatory group are proposed, which better fit with the energetic and environmental parameters. This index allows to compare the proposed system with the reference case. Table 5 shows the indicators chosen.

3. Results and interpretation

3.1. Life cycle assessment

The LCA analysis consists of two phases:

- Evaluation of the climate change impact of the proposed PV-PCM system (proposed scenario).
- Comparison with the traditional fully electric system (base line scenario).

3.1.1. Climate change impact of the proposed system

The environmental impact of the materials used in the integrated panel has been evaluated according to Eq. (2). The impact on climate change considers both the production and the disposal phase, neglecting the life-time (around 20 years), since the system does not need maintenance operations. This means that during the entire life-cycle, the PV panel does not produce emissions and accordingly does not contribute to the climate change impact. Results are shown in Table 6. The zero impact of the disposal phase of the aluminum fins and the copper heat exchanger is due to the possibility of total reuse of these components for further panel manufacturing. It is evident that the production methods of the panel and its end of life are a function of the basic principles of the circular economy: reduce consumption of raw materials, design products with a longer life cycle and finally recycle. Other materials, such as the PMMA box and the PCM, show higher impact as they are intended to the incinerator. Overall, the innovative panel shows a carbon footprint around 105 kg CO_2 eq., including both the material used (Table 6), and the energy consumption for manufacturing (Table 7). More in detail, the environmental burden is due for 40% to manufacturing, where the PMMA production plays a decisive role, causing about the half of production impact. The panel end-of-life causes the 60% of panel life impact. The ease of disassembly and design of the system allows the copper and aluminum to be recovered and integrated to a new panel. The possibility of material recycling has significantly reduced the effect of panel dismantling and the impact is mainly connected with PCM incineration. However, the chemical properties of PCM make it suitable for an energy recovery, and the emissions resulting from incineration (of both PCM and PMMA) are partly balanced by the energetic credit. From this consideration, it is evident that the panel dismantling perfectly fit the principles of European waste pyramid, maximizing the material recycling and favoring an energy recovery, rather than the disposal in landfilling sites for the residue.

Results of the LCA are summarized in Table 8. The first column indicates the energy demand for the DHW service, partly (integrated panel) or fully satisfied by the electrical grid (reference scenario). The second and the third column indicate the carbon footprint estimated for the two scenarios on the energy demand basis, in term of a single day and over twenty years, respectively. It can be noted that the grid provides less energy compared to the reference scenario, as the integrated panel allows a local thermal energy production, indicated in Table 8 as energy saved (fourth column). The remaining share, up to 2.74 kWh/day is provided by the national grid. Overall, the innovative scenario, which implement the innovative panel, shows a carbon footprint of around 8129 kg CO_2 equivalent. This value increases up to 8234.3 kg CO_2 equivalent accounting the impacts of production, operation and disposal

Table 4
Indicators considered for the first index.

Indicator	Description	Numerator	Denominator
1	By-products and/or second raw materials (input) in relation to total material resources (input).	By-products and second raw materials	Total raw material demand
2	Packaging from renewable and/or recycled material resources compared to the total packaging.	Renewable or recycled packaging	Total packaging
3	Waste sent for material recovery in relation to total waste produced.	Recovered waste	Total waste produced

Table 5
Indicators considered for the second index.

Indicator	Description	Numerator	Denominator
1	Total energy consumption	Integrated panel	Reference case
2	Greenhouse gas emissions (GHG in air)	Integrated panel	Reference case

Table 6
CO_2 impact of production and disposal phases.

Material	Weight (kg)	Production impact (kg CO_2)	Disposal impact (kg CO_2)
PCM	4.33	4.76	56.29
Aluminum	0.56	4.95	0.00
Copper	2.39	5.02	0.00
Tin	0.05	0.85	0.00
PMMA	2.56	23.01	7.09
Gasket	0.23	0.72	0.10
Polycarbonate screws	0.08	0.72	0.22
Total		**40.04**	**63.70**

Table 7
CO_2 impact of processes.

Process	Energy of process (kWh)	Climate change (kg CO_2)
Mechanical shear	0.13	0.057
Multi-axis press brake	0.83	0.382
Automatic cutting center	0.67	0.305
Brazing	Negligible	
Injection molding machine	1.33	0.611
CO_2 Laser Cutting	0.14	0.065
Column drill	0.18	0.084
Screwdriver	0.05	0.021
Electric hotplate	0.19	0.085
Total		**1.61**

Table 8
Comparison between environmental impact of the integrated panel and the reference scenario.

Reference case	Energy demand from grid (kWh/day)	EF 3.0 Climate Change (kg CO2 eq./day)	20 years Climate Change (kg CO2 eq.)	Energy saved (kWh/day)
Summer	2.740	1.255	3061.8	–
Mid-season	2.740	1.255	3061.8	–
Winter	2.740	1.255	3061.8	–
Total			9185.4	
Integrated panel				
Summer	2.090	0.957	2335.6	0.650
Mid-season	2.445	1.120	2731.8	0.295
Winter	2.740	1.255	3061.9	0.000
Total			8234.3	

phases.

3.1.2. Comparison with the reference scenario

In the reference scenario, the energy demand for the domestic hot water (DHW) is fully provided by the national electricity grid. As stated in paragraph 2.2, the energy demand in summer, intermediate and winter months has been estimated in 2.74 kWh per day. The analysis has been performed over a period of 20 year and results are summarized in Table 8. The emission factor of the Italian national grid is 0.458 kg CO_2 eq/kWh. The total environmental impact is 9185.4 kg CO_2 equivalent.

The LCA analysis returns an environmental saving of around 950 kg of CO_2, comparing the 9185.4 kg CO_2 equivalent of the reference case and the 8234.3 kg CO_2 equivalent of the proposed integrated system. This reduction, over a period of 20 years, corresponds to an impact reduction of 10.3%. This proves that the proposed integrated panel can be a useful action for the reduction of the environmental impact of the sector. As regard the functional unit, in term of 1 kWh, the environmental impact is 0.41 kg CO_2 equivalent. Considering the long life-time of the panel, its manufacturing and dismantling could be excluded from the carbon footprint of innovative scenario. Nevertheless, authors included this aspect within the assessment to prove as the carbon footprint of the panel life (105 kg of CO_2 equivalent) is minimal compared to the possible advantage resulting from the panel installation in an average household. It is worth noting that results are affected by the emission factors of the electrical grid. In general, a worse quality of the energy mix will return a higher value of the emission factor and accordingly a higher environmental impact, and vice versa. To better investigate the possible correlation, a sensitivity analysis has been performed by varying the electricity mix. Table 9 reports the same comparison of the previous table, between reference case and proposed integrated scenario, for the different energy mixes considered (Italy, Spain, Greece and average European Union). Results show that the percentage variation between the best energy mix (Spain) and the worst one (Greece) is 0.8%.

3.2. Circularity analysis

The quantification of the circularity index, by Italian standard UNI 1608856, adapted to the research context, allowed the evaluation of the circularity of the materials involved in the proposed innovative system. In the first indicator, the aluminum fins and the copper heat exchanger have been considered in the numerator as recyclable materials, while the remaining ones in the denominator. The first indicator results 0.29.

Table 9
Comparison between different energy mixes.

	Reference case 20 years Climate Change (kg CO_2 eq.)	Proposed system 20 years Climate Change (kg CO_2 eq.)	Saving (%)
Italy	9185.6	8234.6	10.3%
Spain	7220.1	6494.8	10.0%
Greece	15703.7	14004.3	10.8%
EU	7460.8	6707.8	10.1%

For the second indicator, to simulate a real application of the innovative panel, a possible packaging has been hypothesized. Indeed, a representative PV-PCM module can be enclosed in 1 cm of polystyrene to avoid damages during the transport and a thin sheet (2 mm) of recycled cardboard. Although it is the most realistic assumption, polystyrene has not been considered recycled and the cardboard is the only material that can be considered as renewable packaging. This aspect is translated into the second indicator results 0.36, which deserves attention considering the current plastic issue. In this regard, more sustainable packaging can be hypothesized or the possible collection by the panel distributor for the possible reuse. The third indicator considers the recovered waste. In this case the polystyrene and the cardboard are recovered waste, while PMMA, gasket and PCM are intended to be disposed with a resulting value of 0.05. This index is not able to consider the possible energetic enhancement of this fractions. The whole index, calculated as average value of the three indicators, is 0.23, that means a material circularity of 31%. Results are summarized in Table 10.

The second indicator concerns the energetic and environmental comparisons between reference scenario and the proposed system. The first indicator considers the total energy consumption for the domestic hot water energy demand, as ratio between the energy saving obtained with the proposed system and the total energy demand. The first indication returns 0.14. The second indicator concerns the environmental impact saving through the proposed system. The ratio turns out to be 0.11. Globally the second index for the performance evaluation shows a 12.8% of circularity. Results are summarized in Table 11.

The use of the new standard highlights the main issues of the innovative system to improve in order to reach the circularity targets. Both indexes shift the attention towards the necessity of a more accurate choice of the materials. In this regard, the choice of a fully recyclable (or reusable) packaging could increase the indicator 2 of the first index up to 1 (hypothesizing the same weight). Additional improvements should be focused on the innovative system itself, mainly on PMMA which could be replaced by a more sustainable materials, the PMMA in fact has been selected only for scientific purpose. Indeed, this material had been chosen as box given its transparency, which results very useful during the experimental tests to visually monitor the solid-liquid phase change of the PCM. This further improvement of the system design could be translated into the increase of the third indicator of the first index up to 0.07. These hypotheses confirm the possibility of the increase of circularity level of the innovative panel with a whole first index of 45.6%.

Table 10
Summary for the first index.

Indicator	Description	Numerator	Denominator	Value
1	By-products and/or second raw materials	3.09	10.59	0.292
2	Renewable or recycled packaging	0.14	0.39	0.363
3	Recovered waste	0.39	7.59	0.05
Average				0.235

Table 11
Summary for the second index.

Indicator	Description	Numerator	Denominator	Value
1	Total energy consumption	124.4	875.5	0.142
2	Greenhouse gas emissions (GHG in air)	1056.3	9185.4	0.115
Average				0.128

4. Conclusions

The present paper represents an excellent example of implementation of sustainability principles, proposing a PV-PCM integrated panel. The system is designed for recycling, able to decrease the dependency from the electric grid, and accordingly its share of non-renewable energy. The combination of sustainability analysis (by LCA and UNI 1608856) with technical aspects allowed the optimization of the panel considering all the aspects of its life (from the manufacturing to the end-of-life), often neglected. The LCA translated the possibility of the production of thermal energy for the domestic hot water of a typical residential building by PV-PCM system into a decrease of burden on climate change of 10.3% (compared to the traditional Italian grid mix). The further circularity assessment proved the possibility to improve the circularity of the product up to 45.6%, highlighting sustainability aspects, often neglected within the research field.

CRediT authorship contribution statement

Daniele Colarossi: Conceptualization, Methodology, Validation, Investigation, Resources, Writing, Visualization, Supervision. **Eleonora Tagliolini:** Conceptualization, Methodology, Validation, Investigation, Resources, Writing, Visualization, Supervision. **Alessia Amato:** Conceptualization, Methodology, Validation, Investigation, Resources, Writing, Visualization, Supervision. **Paolo Principi:** Conceptualization, Methodology, Validation, Investigation, Resources, Writing, Visualization, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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